PROJECT DESCRIPTION · AN EXPLAINABLE-AI LABORATORY FOR CHEST RADIOGRAPHS

XRay*Lens* AI

RESEARCH & EDUCATION ONLY
NOT FOR DIAGNOSIS

A pneumonia classifier gives you an answer. XRayLens shows how it got there, how far to trust it, and where it breaks.

01 Problem

Medical-imaging AI is often presented as one impressive accuracy number. That number can come from testing a model on images it was trained on. Saliency heatmaps then look convincing even when the model relies on text markers or image borders. Students, clinicians-in-training and the public rarely get a way to **question** a model's output.

02 Solution

XRayLens is an interactive lab built around a real, publicly released ResNet-50 pneumonia classifier. It turns each prediction into a four-step investigation, and it treats the evaluation as a product feature, not a footnote.

- **Choose** from 8 teaching cases picked by fixed rules from the evaluation, or upload PNG/JPEG/DICOM.
- **Read** the label, both scores, logit margin and saturation, alongside how 400 external adult images scored.
- **Explain** with Grad-CAM for either class. Heat on lungs, image frame and padding is always shown against area share. A live **randomisation sanity check** (Adebayo et al. 2018) shows whether the map depends on the learned weights.
- **Stress-test** with 8 anatomy-preserving edits (crop, noise, burned-in marker, inversion…), re-scored live.
- **Reliability dashboard**: leakage comparison, live threshold explorer, ROC, calibration, robustness, limitations.

03 Technical approach

Python 3.12 · PyTorch · Streamlit · scikit-learn · pydicom, plus a static web build running the same model in the browser with ONNX Runtime Web (bit-exact Pillow preprocessing, scores within 5×10^{-6} of PyTorch). Model: *Purino/resnet50-chest-xray-pneumonia*, pinned to a Hub commit, with preprocessing **tested for exact parity** with the model card's reference code. Grad-CAM is implemented from scratch with hooks on `layer4` and back-propagates the **logit difference**, because this model's softmax saturates and softmax gradients vanish. 20 automated tests cover preprocessing, input handling, Grad-CAM properties and metric correctness. Every dataset is pinned and sampled with a fixed seed.

04 Evaluation: 1,462 images, 4 cohorts

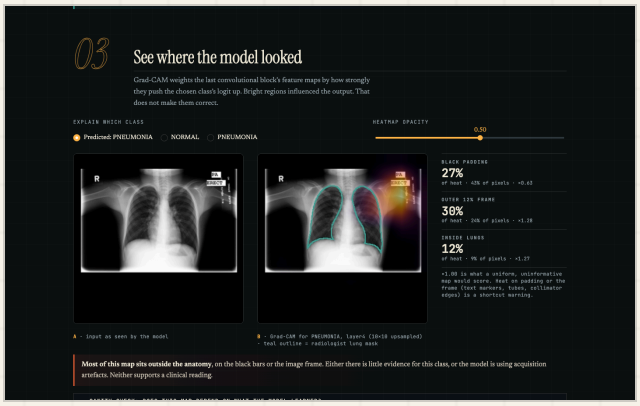
Kermany test · n=624 · seen in training

~~1.000~~ AUC

RSNA adults · external · n=400

38.5% specificity

Finding	Measured
RSNA sensitivity / specificity	99.5% (97.2–99.9) / 38.5% (32.0–45.4)
RSNA AUC (bootstrap 95% CI)	0.913 (0.883–0.939)
Normal adults called pneumonia	RSNA 62% · Montgomery 53% · Shenzhen 20%
Calibration (RSNA, ECE)	0.31; 76% of outputs <0.001 or >0.999
Burned-in marker flips (n=838)	91 → PNEUMONIA · 0 → NORMAL
Grad-CAM in lungs (Montgomery masks)	27% of heat on 21% of pixels (×1.24)



A healthy adult scored 0.997 PNEUMONIA. The Grad-CAM peak sits on the "PA ERECT" marker, and cropping the frame away turns the call to NORMAL (0.10).

05 Limitations

No valid in-distribution test exists, because the published weights were refit on every Kermany image. External cohorts are adults, so population shift and generalisation error are confounded. RSNA "lung opacity" is not confirmed pneumonia. Samples are modest, so all metrics carry 95% CIs. Grad-CAM is a coarse view of one layer. **Not clinically validated, and not a medical device.**

Live demo: xraylens-ai.vercel.app (runs in your browser) · Repository: [add GitHub link]

Data: Kermany 2018 (CC BY 4.0) · RSNA 2018 · NLM Montgomery & Shenzhen · Code: MIT